Banking Customer & Operations Analytics

Project report: insights, data model and recommendations



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INSTANT Data Analysis Training | Sprint 2

Tools: Power BI Desktop, Power Query (M), DAX, Excel/CSV

01 OVERVIEW

About the project

An interactive Power BI report that helps bank management understand customer behaviour, account performance, transaction activity, the loan portfolio and customer support quality. It was built as Sprint 2 of the INSTANT Data Analysis Training. The brief was to think like a banking analyst: answer real business questions, not just draw charts.

Business questions

Area	Question
Customers	Who is active, who is drifting away, and who is at risk of leaving?
Accounts	Where does the money sit, and how much of it belongs to inactive customers?
Transactions	Is activity growing? Which transactions look unusual and need a second look?
Loans	How big and how well priced is the portfolio, and what matures next?
Support	Are customer problems actually being solved, and who keeps calling back?

Data

Training dataset provided by the program, six tables. All "days since" and "active / matured" logic uses the as-of date **11 May 2025**, the last date in the data.

Table	Rows	Notes
Customers	5,050	Join date, state
Accounts	5,100	Type, balance, created date
Transactions	20,000	12 May 2022 to 11 May 2025
Loans	2,600	Type, amount, rate, start and end date
Cards	4,000	Type, issued and expiration date
Support calls	3,100	11 May 2024 to 11 May 2025, resolved Yes/No

Key definitions

Term	Definition
Active customer	Account holder with a transaction in the last 90 days
Cooling	Last transaction 91 to 180 days ago
Inactive	Last transaction more than 180 days ago, or no transaction at all
At-risk customer	Inactive and with at least one unresolved support call
High-value transaction	Amount of 9,500 or more (about the top 5%)
Money out	Withdrawals and payments. Transfers are shown separately because the data does not say which way they go
Loan portfolio	Loans that started on or before the as-of date and end after it

02 DATA PREPARATION

Power Query and data quality

- Imported every table with parameters for the folder path and file name.
- Converted text timestamps to real dates with one reusable function.
- Trimmed text, standardised values and set correct data types.
- Removed duplicate keys (none were found).
- Dropped personal data the analysis does not need: email, phone and address. Card numbers are reduced to the last four digits.
- Added helper columns: balance bands, rate bands, loan status, loan term, card status, resolved flag, high-value flag, money direction.
- Built a date table from 2015 to 2035 (loans end as late as 2035) and marked it as the date table.

Data quality findings

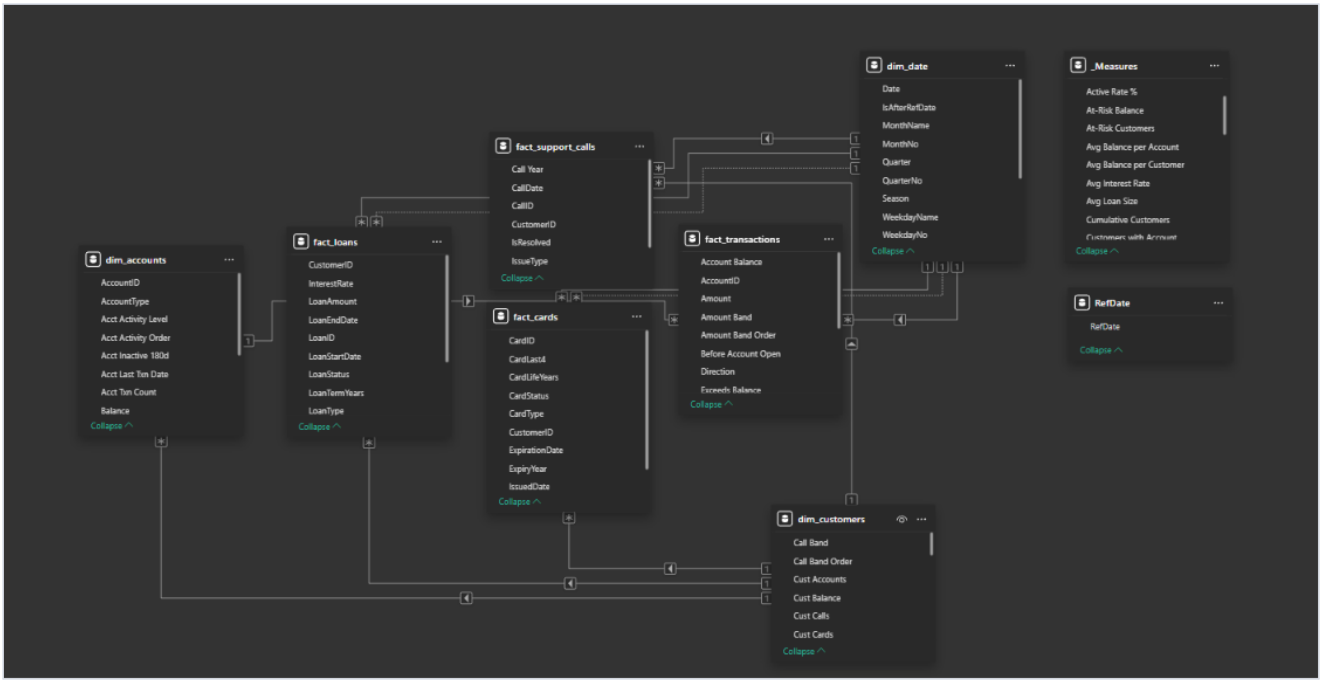
Problems in the source data were flagged and documented rather than silently corrected.

Finding	Count	How it was handled
Nulls, duplicates, orphan keys	0	Verified on every table
Customers with no account	1,851 (37%)	Kept. Shown as "No account"
Customers with no record in any other table	50 (IDs 5001-5050)	Kept. Shown as "No products"
Accounts created before the customer joined	2,488 (49%)	Kept and documented. The correct date is unknown
Transactions dated before the account was opened	3,172 (16%)	Flagged in a column
Withdrawals larger than the account balance	236	Flagged and listed on the Transaction page
Accounts dated after the data date	7	Flagged (IsFutureDated)
Loans starting after the data date	12	Status "Not started", excluded from the active portfolio
State recorded as "Army"	531	Renamed "Military (APO/AE)"

03 DATA MODEL

Star / snowflake schema

Customers and accounts are the dimensions. Transactions, loans, cards and support calls are the facts. All relationships are single direction. The date table is deliberately **not** linked to customer join dates or account created dates, because that would create a second path to the fact tables. Join year and join month are columns on the customer table instead.



From (many)	To (one)	Active
fact_transactions[AccountID]	dim_accounts[AccountID]	Yes
dim_accounts[CustomerID]	dim_customers[CustomerID]	Yes
fact_loans / fact_cards / fact_support_calls [CustomerID]	dim_customers[CustomerID]	Yes
fact_transactions[TransactionDate]	dim_date[Date]	Yes
fact_support_calls[CallDate]	dim_date[Date]	Yes
fact_loans[LoanStartDate]	dim_date[Date]	Yes
fact_cards[IssuedDate]	dim_date[Date]	Yes
fact_loans[LoanEndDate]	dim_date[Date]	No (maturity)
fact_cards[ExpirationDate]	dim_date[Date]	No (expiry)

04 DAX MEASURES

Measure library

All measures live in a dedicated _Measures table, grouped by area.

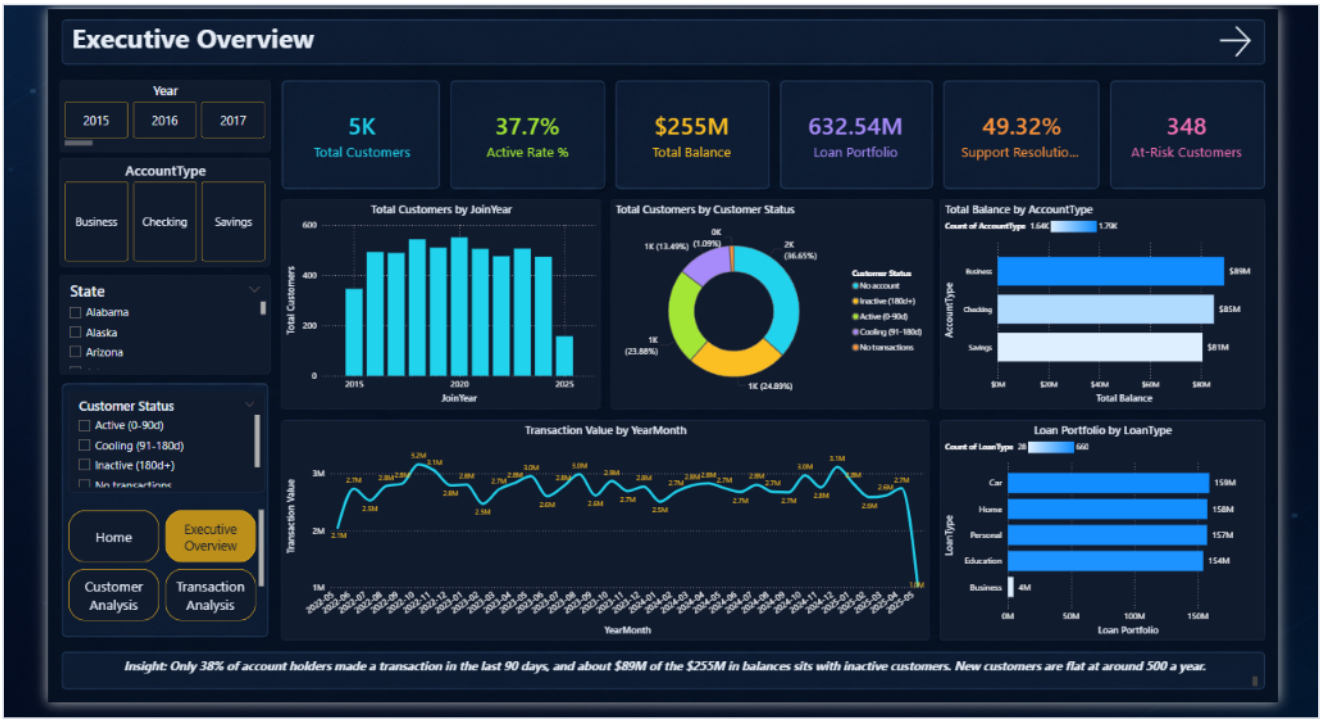
Area	Measures
Customers	Total Customers, Customers with Account, Account Penetration %, Active Customers 90d, Active Rate %, Inactive Customers, At-Risk Customers, Customers without Card, Cumulative Customers
Accounts	Total Accounts, Total Balance, Avg Balance per Account, Avg Balance per Customer, Inactive Balance, At-Risk Balance
Transactions	Total Transactions, Transaction Value, Money In, Money Out, Net Flow, High-Value Txns, Withdrawals above Balance, Txns before Account Open, Value L12M, Value Prior 12M, Value Growth L12M %
Loans	Active Loans, Loan Portfolio, Avg Interest Rate, Avg Loan Size, Maturing Next 12M, High-Rate Share %, Loan Originations
Support	Total Calls, Unresolved Calls, Support Resolution %, Customers Who Called, Repeat Callers, Customers with Unresolved

Customer-level columns

Built on the customer table: number of accounts, total balance, total loan amount, number of cards, number of calls, unresolved calls, last transaction date, days since last transaction, customer status, risk flag, tenure band and product mix.

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Executive Overview



Page insight. Only 38% of account holders made a transaction in the last 90 days, and about $89M of the $255M in balances sits with inactive customers. New customers are flat at around 500 a year.

What to notice

- Headline KPIs: 5K customers, 37.7% active rate, $255M total balance, $632.54M loan portfolio, 49.32% support resolution, 348 at-risk customers.
- Only 63% of customers hold an account; 541 customers hold no product at all.
- Balances are spread almost evenly across Business ($89M), Checking ($85M) and Savings ($81M).
- Monthly transaction value stays near $2.5M to $3.2M with no growth trend. The last point drops because the final month is partial.

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Customer Analysis



Page insight. 1,312 account holders (41%) are inactive and hold $89.5M, and 348 of them also have an unresolved support call. Activity is the same at every tenure, so this is not only a new-customer problem.

What to notice

- Account penetration is 63.3%, with an average balance of $79.6K per customer.
- Active customers hold $113M, cooling $52M and inactive $86M.
- The top 10 at-risk customers table gives the retention team a ready call list with balance, days inactive and unresolved calls.
- Active rate is 35% to 39% across all tenure bands, so tenure does not explain inactivity.
- Military (APO/AE) is the largest state group with 531 customers; the next states have around 100 each.

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Transaction Analysis



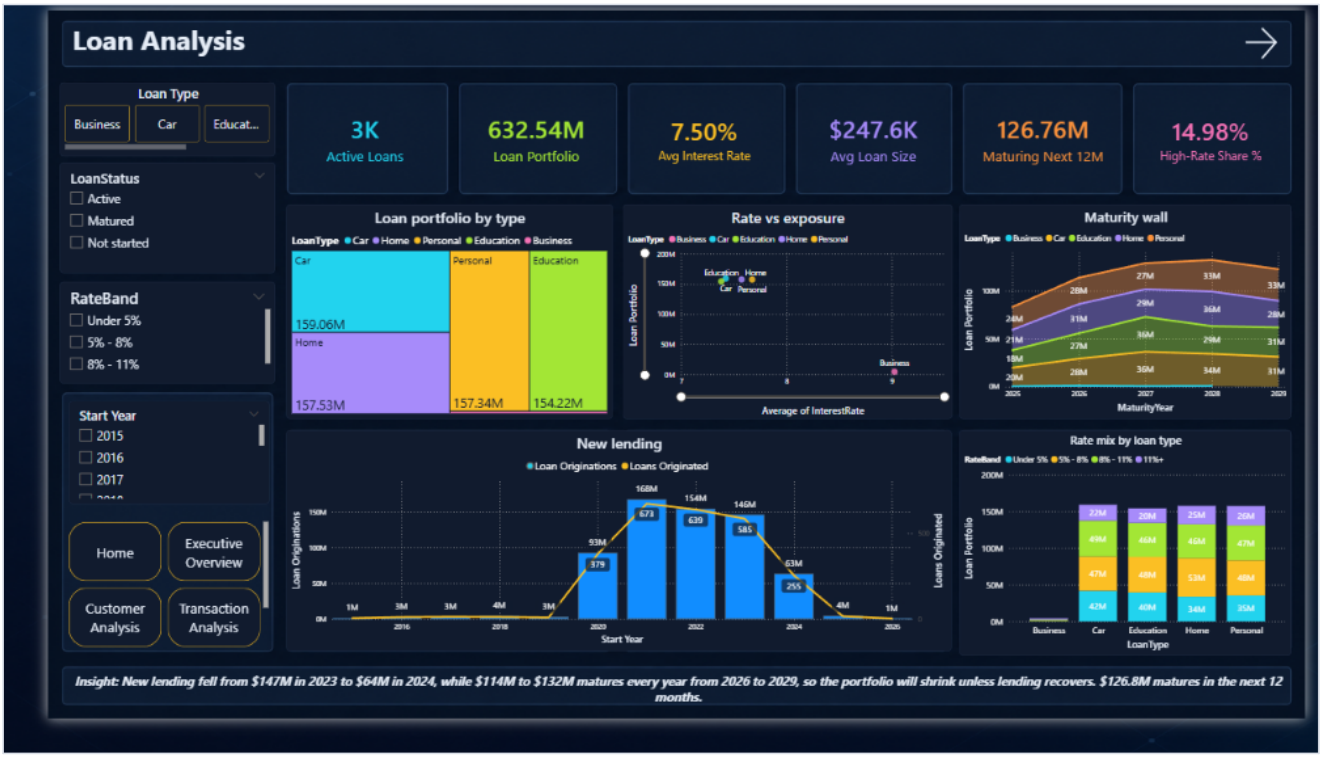
Page insight. Transaction value is flat (+0.8% over the last 12 months) and money out is twice money in. 236 withdrawals are larger than the account balance, and 16% of transactions are dated before the account was opened.

What to notice

- 20K transactions worth $100.11M, with a net flow of -$24.84M (transfers excluded because direction is unknown).
- 1,008 high-value transactions of $9,500 or more.
- Business accounts carry slightly more value ($34M) than Checking and Savings ($33M each).
- The withdrawals-above-balance table lists the top cases for review by a control team.

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Loan Analysis



Page insight. New lending fell from $147M in 2023 to $64M in 2024, while $114M to $132M matures every year from 2026 to 2029, so the portfolio will shrink unless lending recovers. $126.8M matures in the next 12 months.

What to notice

- $632.54M across about 2.5K active loans, with an average interest rate of 7.50% and an average loan size of $247.6K.
- Car ($159M), Home ($158M), Personal ($157M) and Education ($154M) are almost equal in size.
- Rates barely differ by loan type. Business loans are only $4.4M but carry the highest average rate (9.4%).
- 14.98% of the portfolio is priced at 11% or more.

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Customer Support



Page insight. Only 49.3% of 3,100 calls were resolved, and Card Issue is the weakest at 46.5%. 646 customers called more than once, and 1,339 still have an unresolved call.

What to notice

- Resolution is far below the 70% target line (the target is an assumption, because the data has no SLA).
- Issue types are almost evenly split: Transaction Dispute 25.9%, Account Access 25.5%, Card Issue 24.3%, Loan Query 24.3%.
- Overall resolution stays between 48% and 50% in every quarter, so there is no improvement over time.
- 2.3K customers called at least once; 646 of them called more than once.

10 FINDINGS

Key insights

1	**Only 63% of customers hold an account**, and 541 customers hold no product at all.
2	**Activity is low.** 38% of account holders transacted in the last 90 days. 1,312 account holders are inactive and hold about **$89.5M**, around 35% of all balances.
3	**348 customers are at risk** (inactive with an open complaint), holding about $23.9M.
4	**Growth has stalled.** New customers stay around 500 a year, and transaction value is up only 0.8% over the last 12 months.
5	**Money out is about twice money in** ($49.8M against $25.0M, transfers excluded).
6	**Unusual activity:** 1,008 transactions of $9,500 or more, 236 withdrawals above the account balance, and 16% of transactions dated before their account existed.
7	**The loan portfolio is $632.5M across 2,555 active loans.** New lending fell from $146.5M in 2023 to $63.5M in 2024, while $114M to $132M matures every year from 2026 to 2029. About **$126.8M matures in the next 12 months**.
8	**Pricing does not follow loan type.** Average rates sit between 7.3% and 7.7% for the four large loan types, and 15% of the portfolio is priced at 11% or more. Business loans are only $4.4M but carry the highest average rate (9.4%).
9	**Borrowers hold little else with the bank.** 36% of borrowers have no account, and most have loans larger than their deposits.
10	**Support resolves only 49.3% of calls.** Card issues are the weakest (46.5%). 646 customers called more than once and 1,339 still have an unresolved call.

11 ACTIONS

Recommendations

1	**Win back inactive customers** Start with the 348 at-risk customers, then the high-balance inactive ones, with a personal call and a clear offer.
2	**Fix first-contact resolution** Set a resolution target, start recording resolution time, and review the card-issue process first. Close open complaints for customers who also hold loans.
3	**Plan the maturity wall** Contact customers whose loans end in the next 12 months with refinancing offers, and restart new lending so the portfolio does not shrink.
4	**Review loan pricing** Check why rates barely differ by loan type and why 15% of the book sits above 11%. Look at whether the small business segment deserves more attention.
5	**Cross-sell** Offer cards to the 1,426 account holders without one, and offer accounts to borrowers who have none.
6	**Strengthen controls** Review withdrawals above balance and transactions of $9,500 and above, and fix the source of transaction and account date mismatches.
7	**Improve data quality** Validate dates at entry and store state values properly.

Assumptions and limitations

- The as-of date is 11 May 2025. The first and last months of transactions and calls are partial and are excluded from monthly charts.
- Support calls only carry a resolved Yes/No flag, so resolution time and SLA cannot be measured. The 70% target on the gauge is an assumption.
- Transfer direction is not in the data.
- Card data is modelled and used for product mix and cross-sell, but the report has no dedicated cards page. Card issuance and expiry trends are a natural next step.
- The conflicting dates between customers, accounts and transactions were kept and flagged rather than corrected.

Challenges

- Date model: linking the date table to join and created dates created ambiguous paths, so only the fact dates are linked and the others are used through columns.
- Sorting text labels in order (tenure, product mix, bands) caused circular dependencies until the sort columns were built from the original numbers.
- Partial first and last months distorted trend charts and needed filtering.
- Inconsistent dates between tables (accounts before customers, transactions before accounts).
- No resolution time in the support data.

Full project, data and Power Query scripts: github.com/SherifAzb/Banking-Customer-Operations-Analytics